Fall 2025

ECE 216A Homework 2

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Question 1

a.

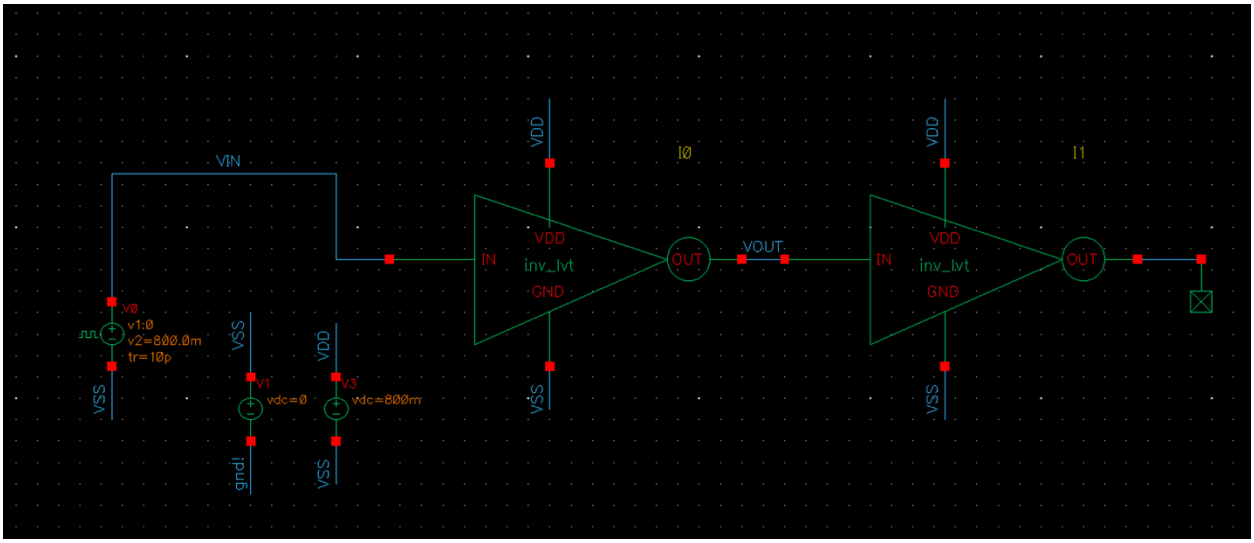


Figure 1. Schematic for FO-1 Inverter

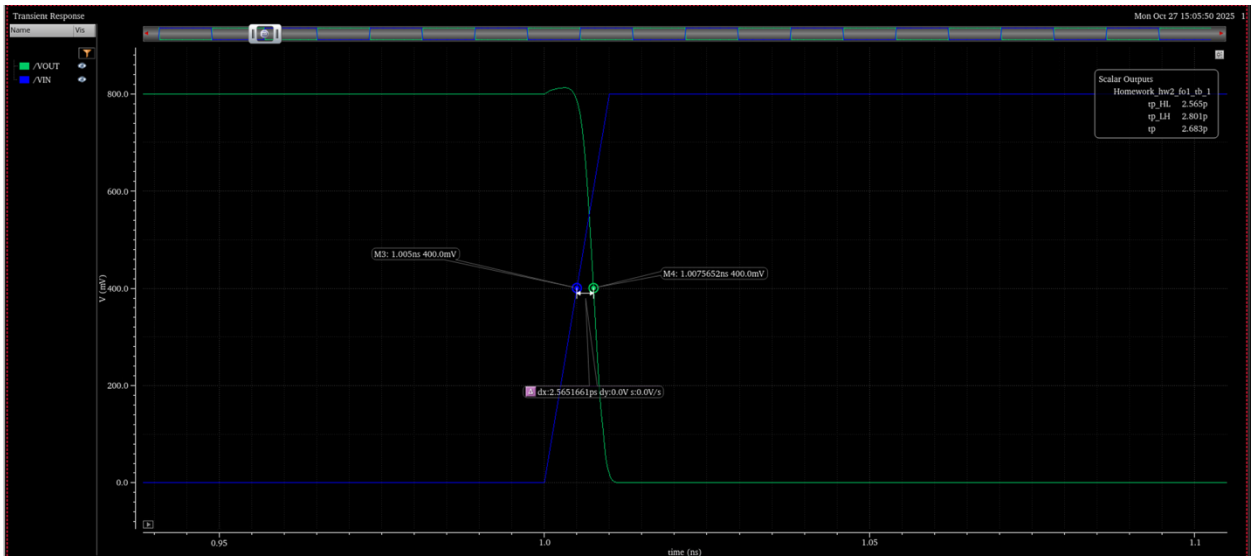


Figure 2. Simulation Results for FO-1 Inverter tp_HL

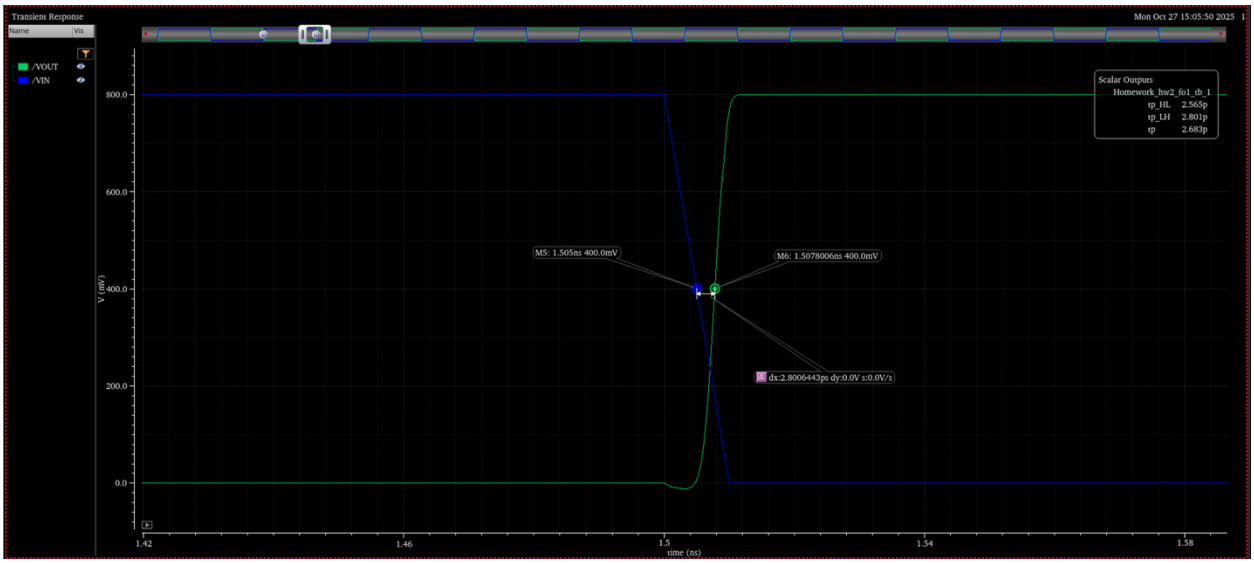


Figure 3. Simulation Results for FO-1 Inverter tp_LH

From simulation results, tp of FO-1 inverter is 2.683 ps.

b.

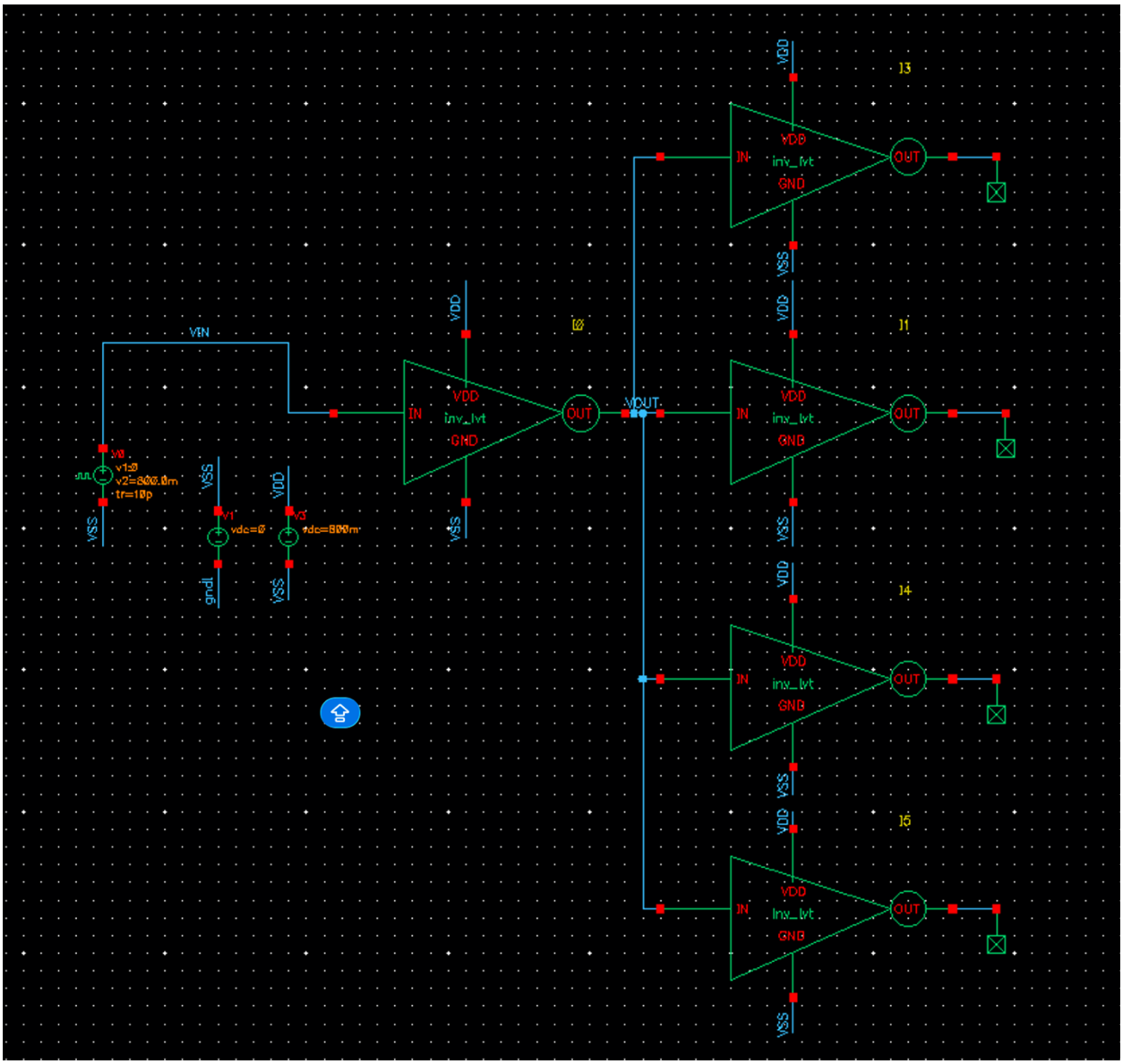

VIN
VDD
VSS
VOUT
IN
OUT
inv_lvt
GND
I0
I1
I3
I4
I5
V0
V1
V3
vdc=0
gnd!

Figure 4. Schematic for FO-4 Inverter

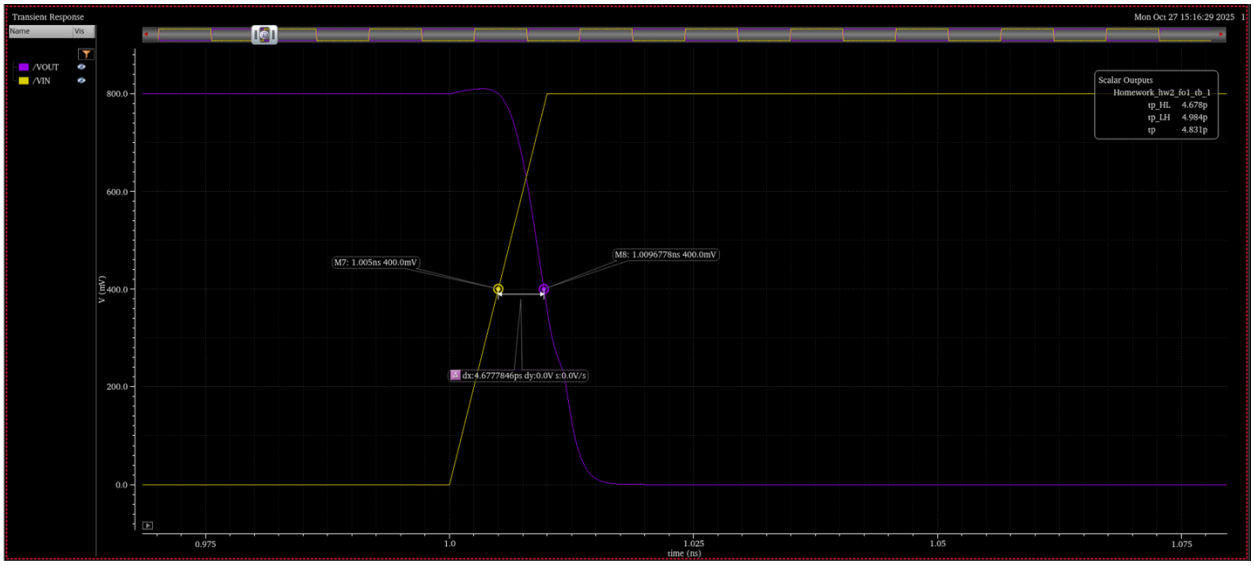


Figure 5. Simulation Results for FO-4 Inverter tp_HL

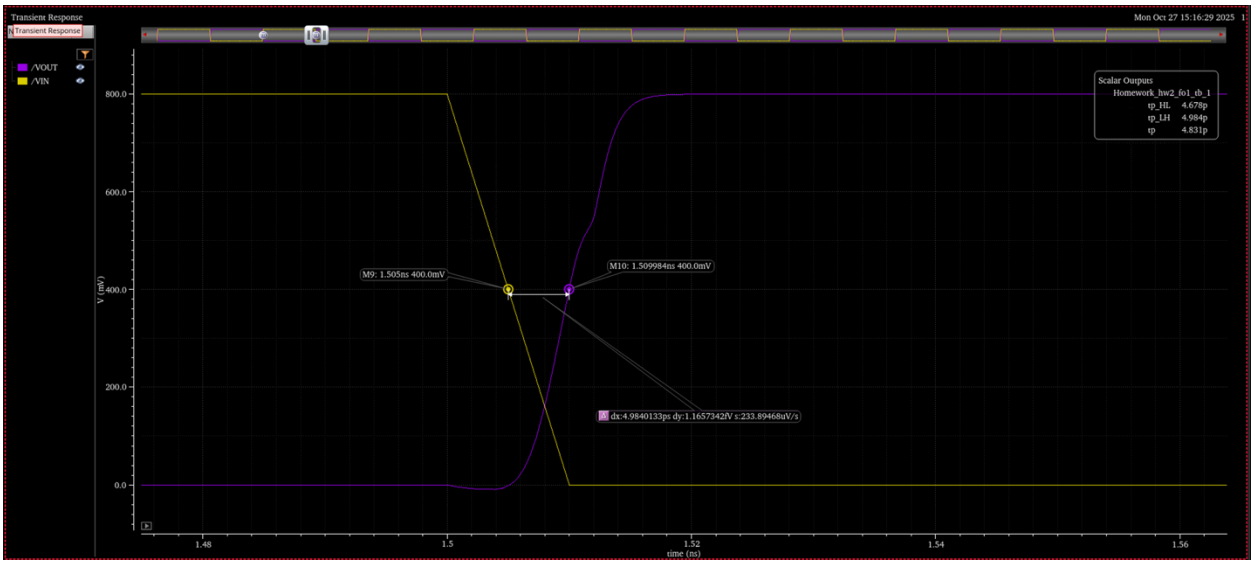


Figure 6. Simulation Results for FO-4 Inverter tp_LH

From simulation results, tp of FO-4 inverter is 4.831 ps.

c.

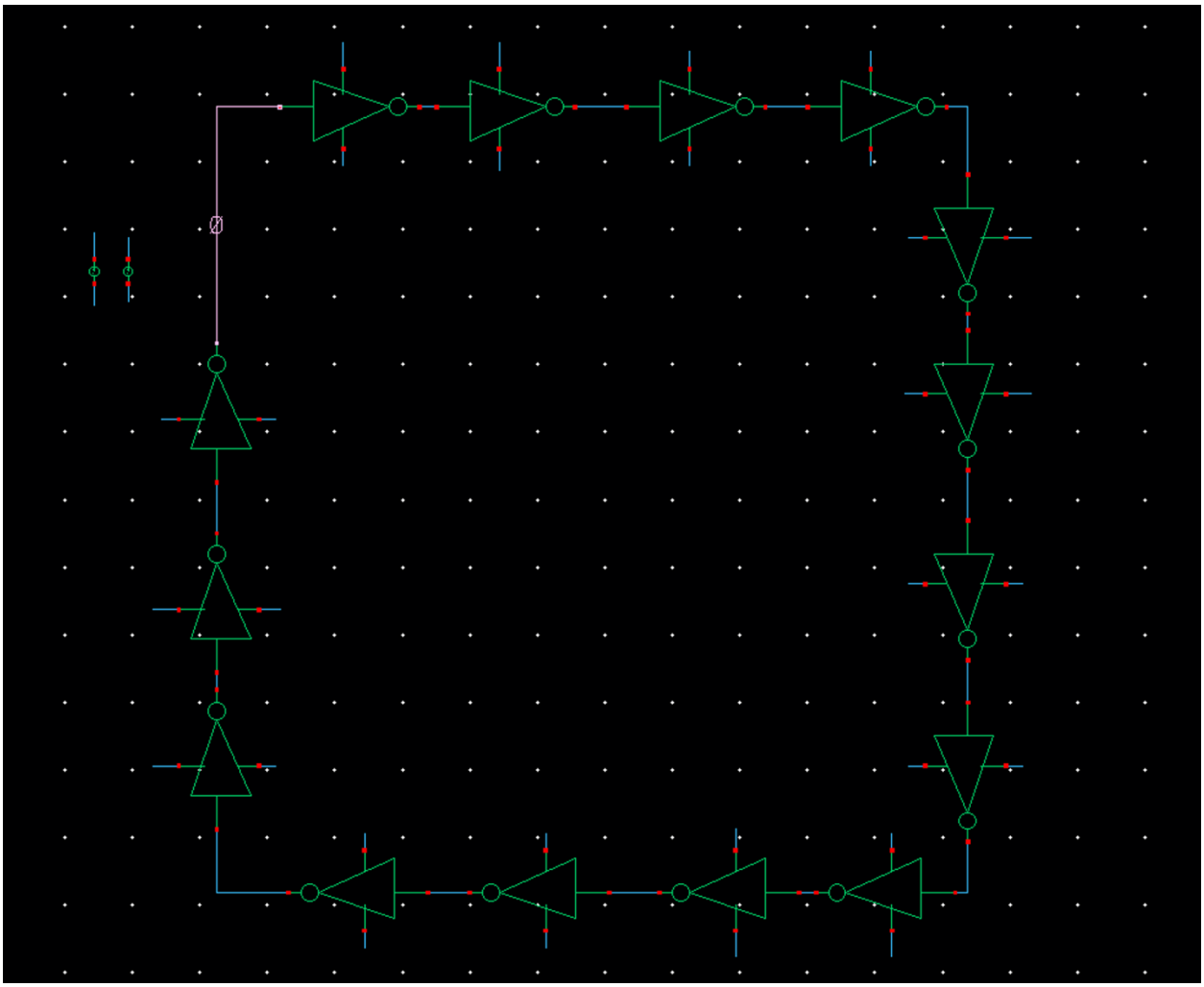


Figure 7. Schematic for 15-Stage Ring Oscillator

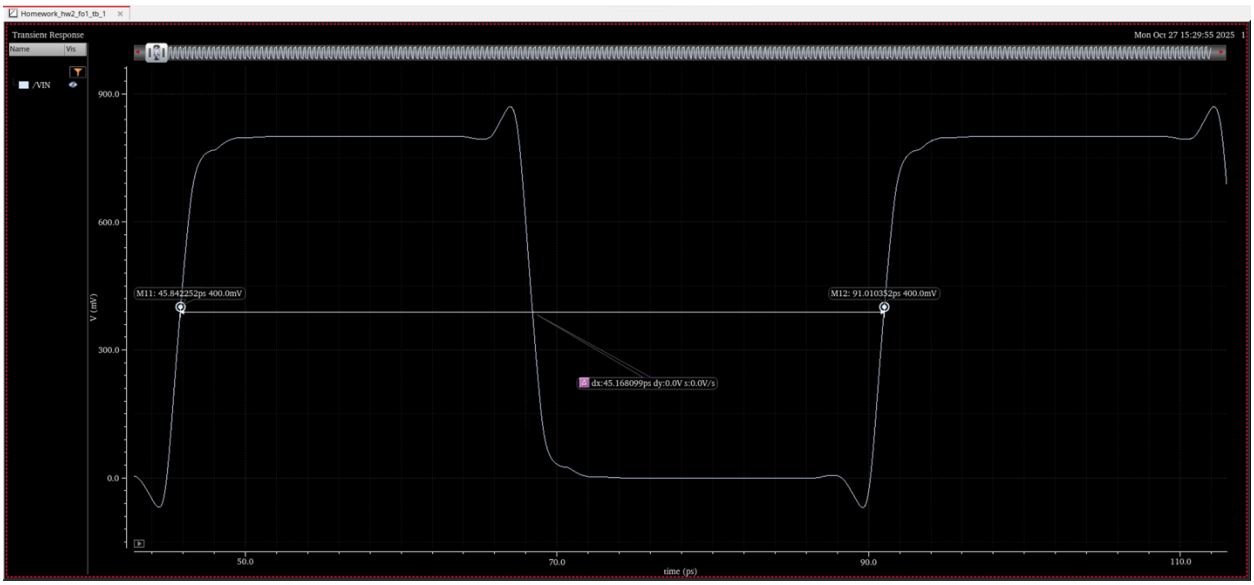


Figure 8. Simulation Results for 15-Stage Ring Oscillator Period

From simulation results, period of 15-stage ring oscillator is 45.168 ps.

1. d

From simulation of the FO-1 invertor,

$$t_p = 2.683\ ps$$

$$\frac{t_{FO-4}}{t_{FO-1}} = \frac{g_{inv} h_{FO-4} + \gamma p_{inv}}{g_{inv} h_{FO-1} + \gamma p_{inv}} = \frac{1 \times 4 + \gamma \times 1}{1 \times 1 + \gamma \times 1} = \frac{4+\gamma}{1+\gamma}$$

$$= \frac{5}{2}$$

$$= 2.5$$

$$t_{FO-4} = 3 t_{FO-1} = 6.7\ ps$$

$$f_{osc} = \frac{1}{2 \times 15 t_{FO-1}} = \frac{1}{30 t_{FO-1}} = 12.42\ GHz$$

2.

a. $f_{opt} = \sqrt[3]{GBH} = \sqrt[3]{(1\times1\times1)(1\times1\times1)\times\frac{64}{1}} = 4$

3nd Stage : $f = gbh = 1\times1\times\frac{64}{y} = 4$

$y = 16$

2nd stage : $f = gbh = 1\times1\times\frac{16}{x} = 4$

$x = 4$

b. Path Delay $= 3f_{opt} + \gamma(p_1 + p_2 + p_3)$

$= 3\times4 + 0.5\ (1+1+1)$

$= 13.5$

c. Path Delay $= g_1b_1h_1 + g_2b_2h_2 + g_3b_3h_3 + \gamma(p_1+p_2+p_3)$

$= x + \frac{32+y}{y}\cdot\frac{y}{x} + \frac{32}{y} + 3\gamma$

$= x + \frac{32+y}{x} + \frac{32}{y} + 3\gamma$

$\frac{\partial \text{ Path Delay}}{\partial x} = 1 - \frac{32+y}{x^2} = 0$

$x = \sqrt{y+32} \quad \cdots ①$

$$\frac{\partial \text{Path Delay}}{\partial y} = \frac{1}{x} - \frac{32}{y^2} = 0$$

$$x = \frac{y^2}{32} \quad \cdots ②$$

①, ② $\Rightarrow$ $\begin{cases} x = 6.8 \\ y = 14.8 \end{cases}$

to make x, y integer,

$$\begin{cases} x = 7 \\ y = 15 \end{cases}$$

d. Path Delay $= x + \frac{32+y}{x} + \frac{32}{y} + 3\gamma$

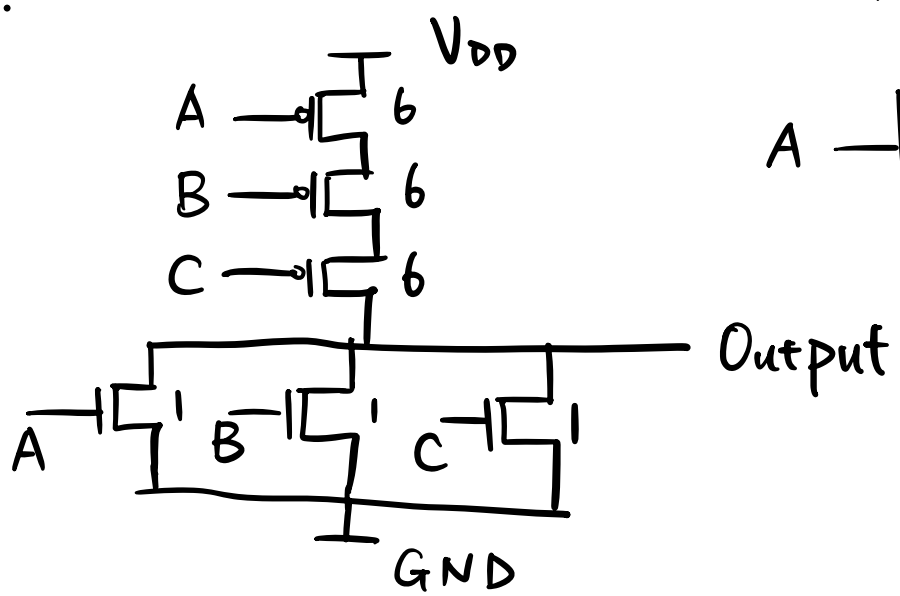
$= 7 + \frac{32+15}{7} + \frac{32}{15} + 3 \times 0.5$

$= 17.35$

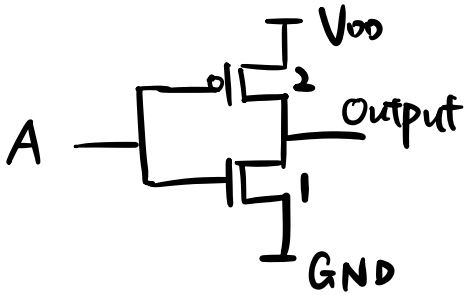
3.

a.

NOR-3



Reference Inv



$$g_{NOR3} = \frac{6+1}{2+1} = \frac{7}{3}$$

$$P_{NOR3} = \frac{6+1+1+1}{3} = 3$$

b. NAND-2 : $g = \frac{4}{3}$, $P = 2$

inverter : $g = 1$, $P = 1$

NOR-3 : $g = \frac{7}{3}$, $P = 3$

$$G = \frac{4}{3} \times 1 \times \frac{7}{3} \times 1 = \frac{28}{9}$$

$$B = 2 \times 1 \times 1 \times 1 = 2$$

$$C_{in} = g\,C_{inv} = \frac{4}{3}C_{inv}$$

$$H = \frac{56\,C_{inv}}{\frac{4}{3}C_{inv}} = 42$$

$$F = GBH = \frac{28}{9} \times 2 \times 42 = 261.33$$

$$P = 2 + 1 + 3 + 1 = 7$$

$$f_{opt} = \sqrt[4]{F} = 4$$

$$D = 4 f_{opt} + P = 4 \times 4 + 7 = 23$$

C.

4th Stage :

$$f_{opt} = gbh = 1 \times 1 \times \frac{56 C_{inv}}{C_{in,4th}} = 4$$

$$C_{in,4th} = 14 C_{inv}$$

$$z\,C_{inv} = 14 C_{inv}$$

$$z = 14$$

3rd Stage:

$$f_{opt} = gbh = \frac{7}{3} \times 1 \times \frac{14 C_{inv}}{C_{in,3rd}} = 4$$

$$C_{in,3rd} = \frac{49}{6} C_{inv}$$

$$y \cdot \frac{7}{3} C_{inv} = \frac{49}{6} C_{inv}$$

$$y = 3.5$$

2nd Stage:

$$f_{opt} = gbh = 1 \times 1 \times \frac{\frac{49}{6} C_{inv}}{C_{in,2nd}} = 4$$

$$C_{in,2nd} = \frac{49}{24} C_{inv}$$

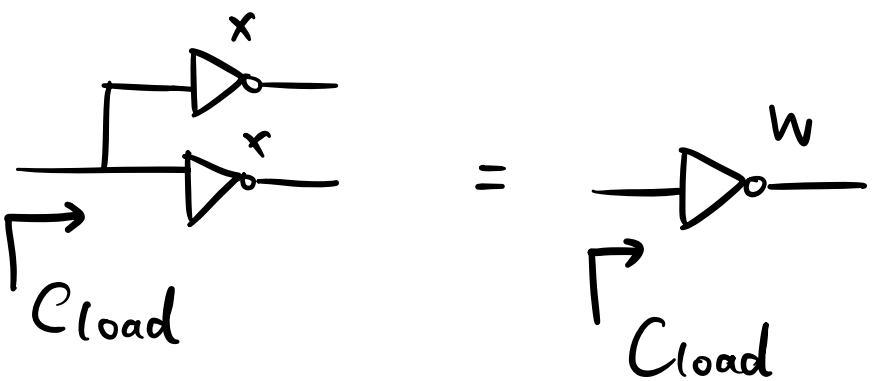
$$x C_{inv} = \frac{49}{24} C_{inv}$$

$$x = 2$$

1st Stage:

$$f_{opt} = gbh = \frac{4}{3} \times 2 \times \frac{\frac{49}{24} C_{inv}}{\frac{4}{3} C_{inv}} = 4$$

d. To match the delay from D to Y to A − Y, make the two NAND-2 have the same delay.



$$2x\,C_{inv} = w\,C_{inv}$$

$$w = 2x$$

$$w = 4$$

4.

a. $F = (A + B)C$

b.

C	A	B	Y	F	Probability
0	0	0	0	0	$0.8 \times 0.6 \times 0.65 = 0.312$
0	0	1	1	0	$0.8 \times 0.6 \times 0.35 = 0.168$
0	1	0	1	0	$0.8 \times 0.4 \times 0.65 = 0.208$
0	1	1	1	0	$0.8 \times 0.4 \times 0.35 = 0.112$
1	0	0	0	0	$0.2 \times 0.6 \times 0.65 = 0.078$
1	0	1	1	1	$0.2 \times 0.6 \times 0.35 = 0.042$
1	1	0	1	1	$0.2 \times 0.4 \times 0.65 = 0.052$
1	1	1	1	1	$0.2 \times 0.4 \times 0.35 = 0.028$

$P_{Y,0} = 0.312 + 0.078 = 0.39$

$P_{Y,1} = 1 - 0.39 = 0.61$

$P_{F,1} = 0.042 + 0.052 + 0.028 = 0.122$

$P_{F,0} = 1 - 0.122 = 0.878$

$\alpha_{0\to1,Y} = P_{Y,0} \times P_{Y,1} = 0.2379$

$\alpha_{0\to1,F} = P_{F,0} \times P_{F,1} = 0.107116$

For C_Y, it can be charged up to V_{DD}

disarged down to 0

For C_F, it can be charged up to $V_{DD} - V_{THN}$

discharged down to 0

$$P_{dynamic} = P_Y + P_F$$

$$= \alpha_{0\to1,Y} \cdot C_Y \cdot V_{DD}^2 \cdot f + \alpha_{0\to1,F} \cdot C_F \cdot V_{DD}(V_{DD} - V_{THN}) \cdot f$$

$$= 0.2379 \times 10fF \times (1V)^2 \times 1GHz + 0.107116 \times 30fF \times (1V)(1V - 0.25V) \times 1GHz$$

$= 0.478911 \times 10^{-5}\ W$

$= 4.78911\ \mu W$

C. Heat Dissipation :

$$E_{C_Y} = C_Y \cdot V_{DD}^2 = 10fF \cdot (1V)^2 = 10fJ$$

$$E_{C_F} = C_F \cdot V_{DD} (V_{DD} - V_{THN})$$

$$= 30fF \cdot (1V)(1V - 0.25V)$$

$$= 22.5fJ$$